

# BayPass’s local-score outlier-region method

A cross-check against the Stage-1-direct climate/mitotype scan

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## Motivation

BayPass ships its own recommended method for turning per-SNP association evidence into outlier *regions*: `compute.local.scores()` (`baypass_public/utils/baypass_utils.R`), implementing the local-score approach of Fariello et al. (2017) and Bonhomme et al. (2019). This is methodologically different from the region-definition already used throughout `module_manuscript_rho05` (LD-based Stage-2 (rho05) clustering, tested against a 10,000-draw  $\Omega$ -structured null covariate/contrast distribution – the “floor-survivor” test). This document asks two questions: (1) what outlier regions does BayPass’s own method identify in our four Stage-1-direct covariates (PC1, PC2, bio\_winter, mitoC2), and (2) is that method’s built-in significance threshold actually well calibrated for this dataset, checked directly against the same  $\Omega$ -structured nulls used everywhere else in this pipeline.

**Bottom line, stated up front:** the local-score method’s analytic threshold is *not* well calibrated here. A meaningful fraction of pure  $\Omega$ -structured null draws (no causal covariate effect at all) produce multiple “significant” windows using BayPass’s own default settings – for the mitotype contrast, the typical null draw already produces several. This document should be read as a diagnostic exercise and a cautionary demonstration, not as a source of new candidate loci for the main manuscript.

## 1. Method

For each SNP, a score  $S_i = \text{BF}(\text{dB})_i/10 - \xi$  (continuous covariates) or  $S_i = -\log_{10}(p)_i - \xi$  (mitoC2’s C2 contrast  $p$ -value) is computed, where  $\xi > 0$  is a fixed per-marker penalty (default  $\xi = 1$ , used throughout). SNPs with negative BF have no natural  $p$ -value, so BayPass’s own code substitutes a random draw,  $-\log_{10}(U)$ ,  $U \sim \text{Unif}(0, 1)$  – this is the source of a real reproducibility caveat discussed in Section 5. A per-chromosome *Lindley process*  $L_i = \max(0, L_{i-1} + S_i)$  then accumulates evidence along the chromosome, resetting to zero whenever cumulative evidence turns negative – this is what lets neighbouring, individually-weak signals combine into one region without any externally-imposed LD clustering. An analytic significance threshold is derived per chromosome from its length and its marker-to-marker  $p$ -value autocorrelation (exact for  $\xi \in \{1, 2, 3, 4\}$ ;  $\xi = 1$  throughout here), at  $\alpha = 0.01$  (BayPass’s default, `pval.local.score.thres`). Markers with estimated  $\hat{\pi} < 0.2$  or  $> 0.8$  are excluded (`min.maf=0.2`, default). Every excursion of  $L$  above threshold is reported as one significant window (chromosome, start, end, size, #SNPs, and the peak position of both the Lindley score and the original statistic within it).

Run at two resolutions:

- **Full-SNP** (`local_score_regions.R`): every genome-wide SNP individually ( $\sim 1.08\text{M}$  retained after MAF filtering per covariate), using the real full-SNP BayPass scan already computed for the manuscript.
- **Stage-1-cluster** (`local_score_regions_S1units.R`): the 18,361 Stage-1-direct clusters ( $n_{\text{SNPs}} \geq 5$ ) used by Module C, one representative marker per cluster. `min.nsnp` (minimum markers/chromosome to compute a threshold) had to be lowered from BayPass’s default (10,000) to 100 – the largest chromosome has only 1,078 Stage-1 units, so the default would silently exclude every chromosome.

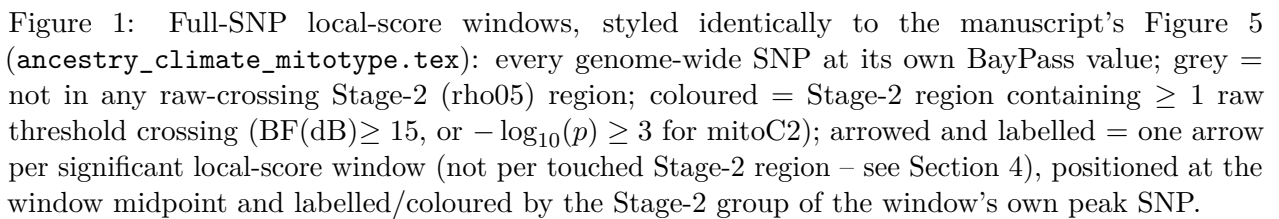
## 2. Observed data, full-SNP resolution

| Covariate  | Raw threshold crossings | Local-score windows | Total window span                                                   |
|------------|-------------------------|---------------------|---------------------------------------------------------------------|
| PC1        | 2,779                   | 6                   | Chr2, Chr8, Chr15 ( $\times 2$ ), Chr24 ( $\times 2$ )              |
| PC2        | 1,484                   | 7                   | Chr4, Chr5, Chr13, Chr24, Chr26 ( $\times 2$ ), Chr27               |
| bio_winter | 2,636                   | 8                   | Chr2, Chr3, Chr8, Chr11, Chr15 ( $\times 2$ ), Chr24 ( $\times 2$ ) |
| mitoC2     | 648                     | 95                  | scattered across 26 of 26 chromosomes                               |

Table 1: Local-score windows, full-SNP resolution, BayPass default parameters ( $\xi = 1$ ,  $\alpha = 0.01$ ,  $\text{MAF} \geq 0.2$ ). Figure 1.

**Striking cross-covariate overlap.** PC1 and bio\_winter’s windows share near-*identical* start coordinates at five of their six/eight loci (Chr2: 10,008,320 bp both; Chr8: 6,712,279 bp both; Chr15: both peaks within  $\sim 1.2$  kb; Chr24: both peaks within  $\sim 0$  bp). PC2 additionally has its own Chr24 window (1,764,859–1,843,054) *nested entirely inside* PC1/bio\_winter’s broader Chr24 window (1,309,506–1,894,413/1,928,191). Three unrelated covariates converging on the same loci is not consistent with covariate-specific climate signal – it is exactly the genome-architecture/ancestry-confound pattern already central to `ancestry_climate_mitotype.tex` (Section 6 there), now recovered independently by a completely different region-detection method. **Confirmed directly (Section 6 below): genome-wide, PC1 and bio\_winter’s full-SNP BF vectors correlate at  $r = +0.89$**  (restricted to the Stage-1-representative markers used elsewhere in this document;  $r = +0.90$  genome-wide) – strongly positive, consistent with the raw covariates’ own near-collinearity ( $r = -0.957$  across the 19 populations) and with the shared windows shown here. PC2, which is genuinely a different axis (raw correlation with PC1:  $-0.155$ ; with bio\_winter:  $-0.030$ ), shows essentially no such relationship with either ( $r = +0.04$  and  $r = +0.03$ ) – consistent with having no local-score windows in common with PC1 or bio\_winter above. Section 6 also reports a striking, reproducible *negative* version of the PC1/bio\_winter correlation at a different (Stage-1-cluster) resolution, traced there to a real but mechanistically unresolved property of that smaller-scale BayPass run – not a competing finding about PC1 and bio\_winter’s true relationship, which this full-SNP result establishes directly.

**Relationship to the floor-survivor candidates.** Of the 12 floor-survivor loci identified elsewhere in this pipeline (PC1: 1, PC2: 2, bio\_winter: 10), only bio\_winter’s three loci on Chr2 (both units) and Chr24 fall inside a full-SNP local-score window; PC1’s Chr24 floor-survivor sits in the  $\sim 170$  kb gap between two adjacent local-score windows, and neither PC2 floor-survivor is recovered at all. The two methods substantially disagree at the level of individual candidate loci, even though (per the paragraph above) they agree on which broad regions show elevated signal shared across covariates.



### 3. Observed data, Stage-1-cluster resolution

| Covariate  | Local-score windows | vs. full-SNP | Windows (chr) |
|------------|---------------------|--------------|---------------|
| PC1        | 1                   | 6            | Chr24         |
| PC2        | 0                   | 7            | –             |
| bio_winter | 2                   | 8            | Chr2, Chr24   |
| mitoC2     | 8                   | 95           | scattered     |

Table 2: Local-score windows, Stage-1-cluster resolution (18,361 units, min.nsnp=100). Figure 2.

Far fewer windows survive at this coarser resolution – expected, since Stage-1 units are deliberately LD-pruned ( $\rho=0.5$ ) to be much less correlated than neighbouring full SNPs, so the Lindley process has far less to accumulate. PC1’s Chr24 window and bio\_winter’s Chr2 and Chr24 windows correspond to the SAME loci recovered at full-SNP resolution and to real floor-survivor candidates – the one part of this cross-check where local-score, Stage-1, and the structured-null floor test all agree. PC2 loses its signal entirely at this resolution.

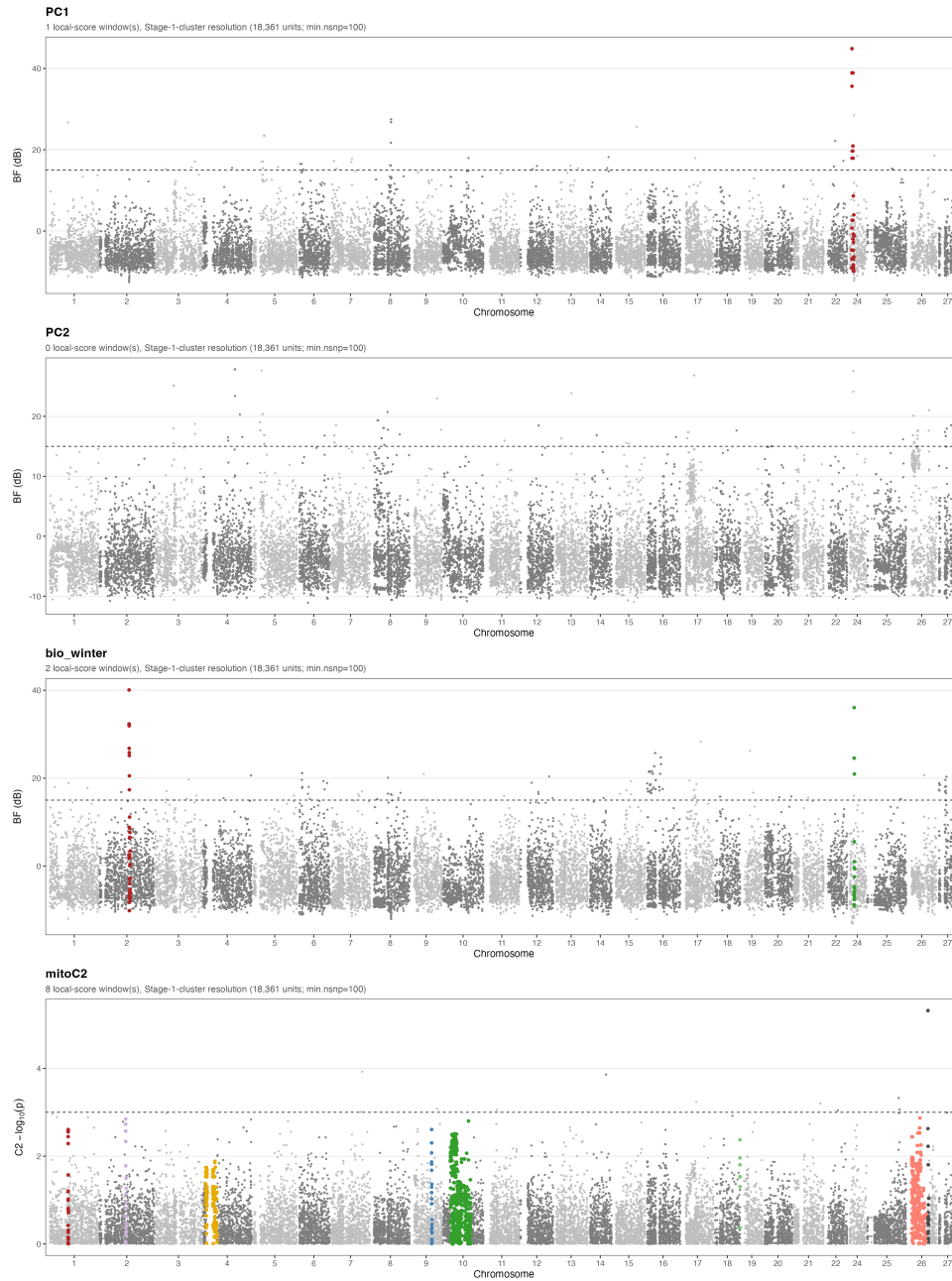


Figure 2: Local-score windows, Stage-1-cluster resolution. Points are coloured by local-score window identity (not Stage-2 group, unlike Figure 1); every genome-wide member SNP of a tested Stage-1 cluster inherits its cluster's own single value (not an independent per-SNP estimate – no such Stage-1-resolution SNP-level scan exists), which is why points inside a coloured window form flat vertical bands rather than the natural scatter in Figure 1.

## 4. Does the analytic threshold survive the $\Omega$ -structured null?

The real question this document exists to answer: if a covariate/contrast carries *no* causal signal – only the populations’  $\Omega$  covariance structure, exactly like the null covariates already built for Module C’s genome-wide calibration (`moduleC_stage1_null_regen.R`) and the mitotype floor test (`moduleB_stage1_mitoC2_null.R`) – how often does BayPass’s own analytic threshold still call something “significant”? Run at Stage-1-cluster resolution (the only resolution at which null draws exist at all – generating a full-SNP null would need  $\Omega$ -structured covariates re-run through BayPass on  $\sim 1.1$ M markers each, an order of magnitude beyond anything already computed in this pipeline).

| Null pool (1,000 draws each)                  | Mean windows | Median | Max | % draws $\geq$ real count |
|-----------------------------------------------|--------------|--------|-----|---------------------------|
| Continuous (PC1/PC2/bio_winter’s shared null) | 0.25         | 0      | 3   | 21.6% ( $\geq 1$ )        |
| mitoC2’s dedicated contrast null              | 3.88         | 4      | 16  | 6.6% ( $\geq 8$ )         |

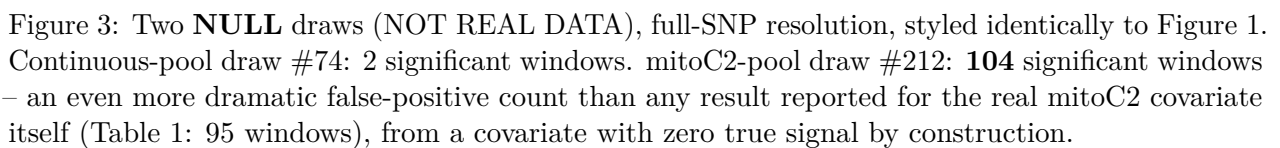
Table 3: Distribution of local-score window counts under 1,000 structured-null draws per pool (`local_score_null_check_S1units.R`), same procedure as the observed data. “% draws  $\geq$  real count” compares against the real covariates’ OWN Stage-1-cluster window count from Section 3 (PC1: 1, bio\_winter: 2, mitoC2: 8 – PC2’s real count is 0, so no comparable tail probability applies).

**Interpretation.** Roughly one in five draws from a covariate with *zero* true signal already produces at least one “significant” local-score window under BayPass’s default threshold; for the mitotype contrast, the typical null draw produces about four. This is a strong, direct demonstration that the analytic threshold under-corrects for this dataset’s population structure – consistent with, and now quantifying, the same conclusion reached independently in `ancestry_climate_mitotype.tex` Section 6 (mitoC2’s genome-wide BF/C2 landscape is indistinguishable from its own  $\Omega$ -structured null on every measure tested there).

## 5. False-positive demonstration

### 5a. Full-SNP resolution – worst of 20 random draws, genuine BayPass pipeline

To avoid any suggestion of cherry-picking the most extreme case out of a large pre-scanned pool, 20 draws were randomly sampled (not the full 1,000) from each null pool, and the *worst* of those 20 (most local-score windows, Stage-1-cluster resolution: continuous draw #74, 1 window; mitoC2 draw #212, 9 windows) was carried forward. Because the nulls are shared across PC1/PC2/bio\_winter, only one continuous-pool draw and one mitoC2-pool draw are needed. Each was then run through the *actual full-SNP BayPass pipeline* – the identical invocation used for the real covariates (same `u_DIEM.geno`, `Omega`, `poolsize`, MCMC settings; only the `-efile/-contrastfile` swapped for the null draw’s raw covariate/contrast values, pulled from the preserved `null_b01.env` row 74 / `nullc2_b02.contrast` row 12) – not the Stage-1-cluster BF inherited across member SNPs used in Section 5b below. This is the first null ever run through full-SNP BayPass in this pipeline (the 10,000-draw scale used elsewhere is infeasible at 1.1M markers) and gives a genuine apples-to-apples comparison with Figure 1.



For context, Figure 4 shows three more high-scoring draws (also from the worst-of-20 samples described above where available) at Stage-1-cluster resolution (member SNPs inherit their cluster’s value, as in Figure 2 – cheaper to generate, illustrating that this is not a one-off result).



strength (Figure 5).

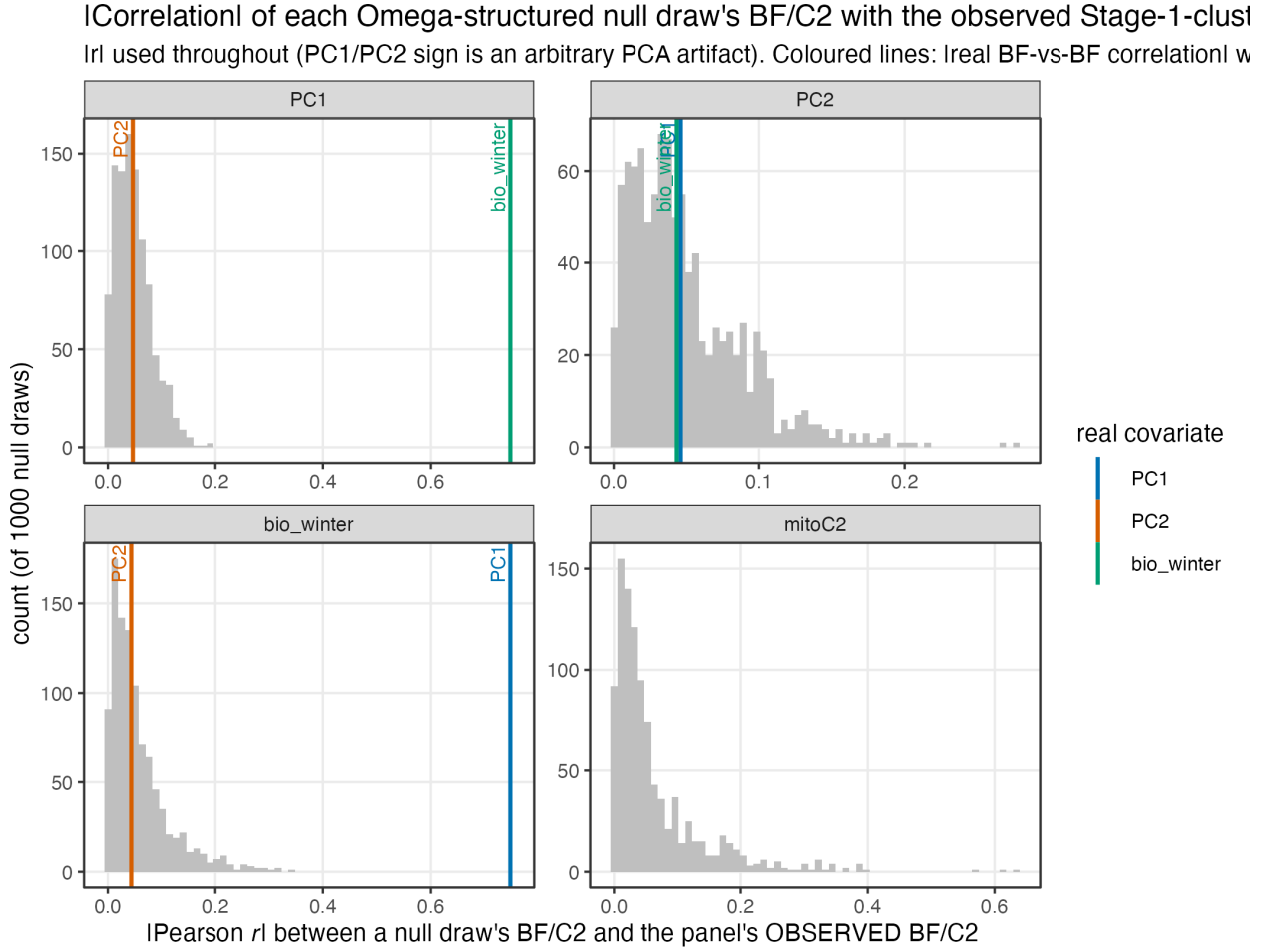


Figure 5: Grey: distribution of  $|r|$  between each of 1,000 null draws' BF/C2 and the observed BF/C2, per covariate, **Stage-1-cluster resolution**. Coloured vertical lines:  $|r|$  the REAL BF-vs-BF correlation between that panel's covariate and another observed covariate, same resolution. Read together with the self-consistency finding below before interpreting the PC1/bio\_winter panels.

**Only PC1 vs. bio\_winter is genuinely extreme at Stage-1-cluster resolution.**  $|r| = 0.748$ , beyond the maximum  $|r|$  seen in any of the 1,000 null draws for either covariate (max 0.190 in PC1's pool, 0.336 in bio\_winter's). PC1 vs. PC2 ( $|r| = 0.046$ ) and PC2 vs. bio\_winter ( $|r| = 0.044$ ) both sit at the 54–58th percentile of their own null  $|r|$  distributions – statistically unremarkable. **But this specific number is resolution-dependent and should not be read at face value** – see below.

**Self-consistency check: does the Stage-1-cluster (S1units) BayPass run agree with the full-SNP run at the same physical markers?** This matters because Stage-1-cluster resolution is the only one with null draws available, so every result above implicitly trusts S1units BF values as “the observed result.” Checked directly: PC1 and bio\_winter's local-score windows sit at near-identical genomic positions (Section 2), which predicts a *positive* BF-vs-BF correlation – yet the number above is strongly *negative*. At full-SNP resolution, restricted to the same 18,361 Stage-1-representative markers, the correlation is  $r = +0.892$  (Table 4) – strongly positive, exactly as predicted by the shared windows and the raw covariates' collinearity. PC1's and PC2's own run-to-run self-consistency (S1units vs. full-SNP BF, same markers) is only  $r = 0.13/0.36$  overall, rising to  $r = 0.89–0.98$  once restricted to loci with any real effect size (**Beta\_is** more than  $\sim 0.05$  from the prior boundary) – i.e. the S1units run is a substantially noisier estimate of the true (full-SNP) result than at first glance,

though individually-reliable at its own strongest loci.

| Covariate  | cor(S1units BF, full-SNP BF), same markers |
|------------|--------------------------------------------|
| PC1        | 0.134                                      |
| PC2        | 0.359                                      |
| bio_winter | 0.451                                      |
| mitoC2     | 0.939                                      |

Table 4: Self-consistency between the two independent BayPass runs, all 18,361 Stage-1-representative markers, unconditional on effect size.

**The PC1/bio\_winter sign reversal at S1units resolution is real and fully reproducible – not random MCMC noise, not a data-alignment bug, and not explained by covariate collinearity – but its mechanism remains unresolved.** Ruled out directly: (i) file identity – u.bio\_winter and the Omega file are byte-identical between the two BayPass run directories; (ii) marker misalignment – MRK order was verified against c15’s row order, and the same physical markers were looked up explicitly; (iii) MCMC convergence failure – pilot-run acceptance-rate diagnostics are near-identical between the PC1 and bio\_winter S1units runs, no anomaly flagged; (iv) random seed – **both covariates’ S1units runs were independently re-run with a different seed (75 vs. the original 74)**. PC1’s own reproducibility is  $r = 0.983$  (orig. vs. reseed) and bio\_winter’s is  $r = 0.968$  – both runs are internally reliable. Using *only* the two independently-reseeded runs (sharing no random state with the original pair, and none with each other’s original run), the anomaly reproduces almost exactly:  $\text{cor}(\text{BF}_{\text{PC1, reseed}}, \text{BF}_{\text{bio\_winter, reseed}}) = -0.751$ , and among the 50 markers with the strongest PC1 effect, **sign(Beta\_is)** agrees for **100% of loci at S1units resolution in both seed pairs**, vs. **0% (perfect disagreement)** for the identical 50 markers at full-SNP resolution ( $r_{\text{full-SNP}} = -0.911$  for that subset, matching the expected negative relationship). This total, seed-independent reversal is a genuine, deterministic property of running BayPass’s covariate-regression model on this specific  $\sim 18\text{K}$ -marker scale for these two covariates – not a fluke, not attributable to any of the usual candidate causes – but *why*  $\Omega$ -correction at this marker-set scale produces it has not been mechanistically resolved here (candidate directions for further work: whether it reflects a real property of the aux/IS grid integration near the  $\beta$  prior boundary interacting with near-collinear covariates, or something specific to how the smaller marker set’s global hyperparameters are estimated) and would need direct examination of BayPass’s internals or consultation with its authors to resolve properly.

**Practical conclusion: trust the full-SNP correlation, not the S1units one, for any claim about PC1’s and bio\_winter’s true relationship.** Figure 1’s shared windows and the full-SNP  $r = +0.89\text{--}0.90$  correlation (Section 2) are the reliable result. The S1units-based  $|r|$  percentiles above remain a valid, self-consistent test of *that specific resolution’s* calibration (Section 4’s window-count conclusion is unaffected, since it compares S1units observed against S1units null on equal footing), but the PC1/bio\_winter correlation specifically should not be read as telling us anything new about those two covariates’ real relationship – Section 2 already established that directly and more reliably.

## 7. QQ-plots: observed vs. pooled null

Figure 6 compares the full distribution of observed BF/C2 values against the pooled distribution from the same 1,000 null draws (quantile-matched), for all four covariates.

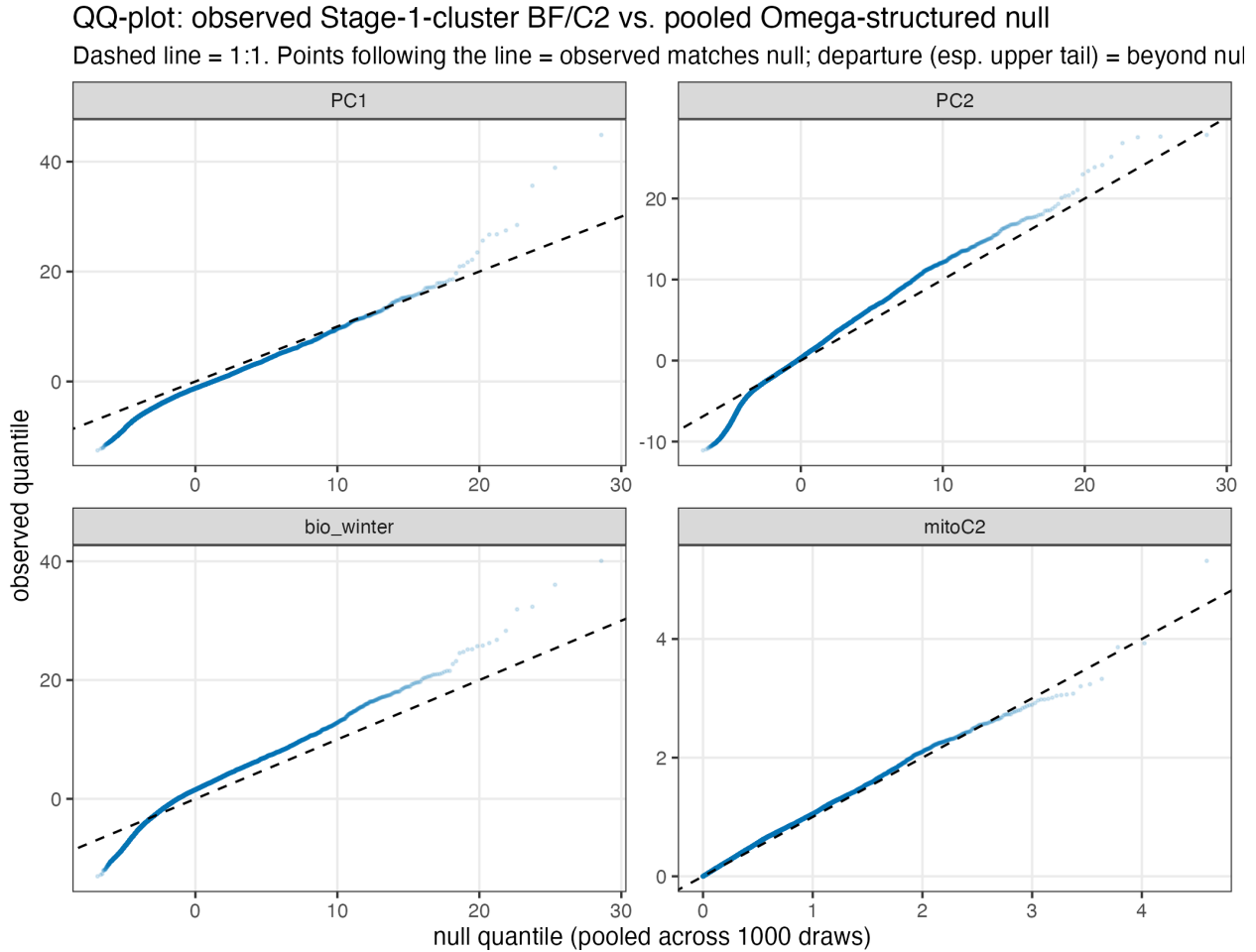


Figure 6: QQ-plots, observed vs. pooled null (1,000 draws), Stage-1- cluster resolution. Dashed line = 1:1.

The same qualitative pattern holds for all four covariates, including mitoC2: the bulk of the distribution tracks the null 1:1 line closely (most Stage-1 units behave exactly like a structure-only covariate would), with clear upward departure concentrated in the upper tail only. This is the expected shape for a genome scan with a small number of real or structurally-inflated outliers sitting on top of an otherwise well-behaved background – it does not by itself distinguish "real candidate loci" from "architecture-driven tail inflation," but it does show that whatever is driving Section 4's window-count miscalibration for mitoC2 is a broad, distributional effect rather than being confined to a few dramatic loci.

## 8. Caveats and limitations

- **The BF-based score is not deterministic.** BayPass's own code draws a random  $p$ -value for every negative BF value (Section 1), so re-running `compute.local.scores()` on the identical BF vector can change the exact window count and boundaries from call to call. All numbers in this document use a fixed seed (`set.seed(1)` in the region-calling scripts; `set.seed(10000+draw)` per panel in the demonstration figure) purely for reproducibility – the instability itself is a genuine, separate calibration concern for continuous covariates, on top of Section 4's finding. mitoC2 (a  $p$ -value statistic) is unaffected and fully deterministic.
- **Arrows are per window, not per touched Stage-2 (rho05) region.** An early version of the full-SNP figure arrowed every Stage-2 cluster overlapped by a significant window and

produced 138–8,055 arrows per covariate, because local-score windows (hundreds of kb to several Mb) routinely span dozens of the much finer-grained ( $\rho=0.5$ , 0.5 cM cap) Stage-2 clusters. That inflated the apparent candidate count and does not reflect what the method actually claims (a WINDOW is the significant unit). Every figure in this document instead places one arrow per window, at its midpoint, labelled/coloured by the Stage-2 group of the window’s own peak SNP – which is itself often not a raw threshold crossing (the method’s entire point is windows where evidence accumulates without any single SNP crossing a naive threshold).

- **No full-SNP-resolution null exists.** Section 4/5’s calibration check and demonstration are necessarily at Stage-1-cluster resolution. The qualitative conclusion (the analytic threshold under-corrects for this dataset) is unlikely to reverse at full-SNP resolution – if anything, denser, more autocorrelated markers give the Lindley process more opportunity to accumulate spurious runs – but this has not been directly tested and would require a large new BayPass campaign to check properly.
- **This document does not replace the floor-survivor test.** The  $\Omega$ -structured-null-calibrated floor-survivor test used throughout `module_manuscript_rho05` remains the primary, trusted candidate-locus method for this project. Local-score is presented here purely as an independent cross-check and a calibration diagnostic, not as a source of new candidates for the manuscript.

## Summary

BayPass’s own local-score outlier-region method, run with its own default settings, recovers a real, directly-confirmed architecture signal: PC1 and `bio_winter`’s full-SNP BF landscapes converge on near-identical genomic windows and correlate at  $r \approx +0.90$  genome-wide (Section 2), consistent with their near-collinearity as raw covariates ( $r = -0.957$ ) and reinforcing the manuscript’s ancestry-confound narrative. PC2 – genuinely a different axis – shows neither shared windows nor a meaningful correlation with either. A direct check against the same  $\Omega$ -structured nulls used everywhere else in this pipeline shows the method’s analytic significance threshold is not well calibrated for this dataset: roughly one in five structure-only null draws for the continuous covariates, and the clear majority for `mitoC2`’s contrast, already produce one or more “significant” windows by chance (Section 4), and a worst-of-20-random-draws null run through the actual full-SNP BayPass pipeline produced 104 significant windows for a covariate with zero true signal by construction (Section 5) – more than the real `mitoC2` covariate itself. Investigating an unexpected negative correlation between PC1 and `bio_winter` specifically at Stage-1-cluster resolution (Section 6) turned up a separate, striking finding in its own right: that correlation is genuine, fully reproducible across independent MCMC seeds, and not attributable to any data or alignment error, yet is the *opposite* sign of the (reliable) full-SNP result for the exact same markers – an unresolved, resolution-dependent property of BayPass’s covariate-regression model at this smaller marker-set scale that is worth flagging to the BayPass developers or investigating further, independent of this document’s main purpose. Combined with a separate, unrelated instability in how the local-score method scores negative Bayes factors (Section 1/8), this method should be treated as a supplementary, exploratory cross-check – informative for the architecture question it incidentally answers, but not a substitute for, or independent confirmation of, the structured-null floor-survivor candidates already reported in the main manuscript.

*References:* Fariello M.-I. et al. (2017) *Mol. Ecol.* 26:3700–3714 – local score method. Bonhomme M. et al. (2019) *Heredity* 123:1–14 – extension to  $\xi = 3, 4$ . Olazcuaga L. et al. (2020) – BayPass C2 contrast statistic.